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Generative and unsupervised DL methods: Generative Adversarial Networks (GANs)

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VAE vs GAN images

VAE

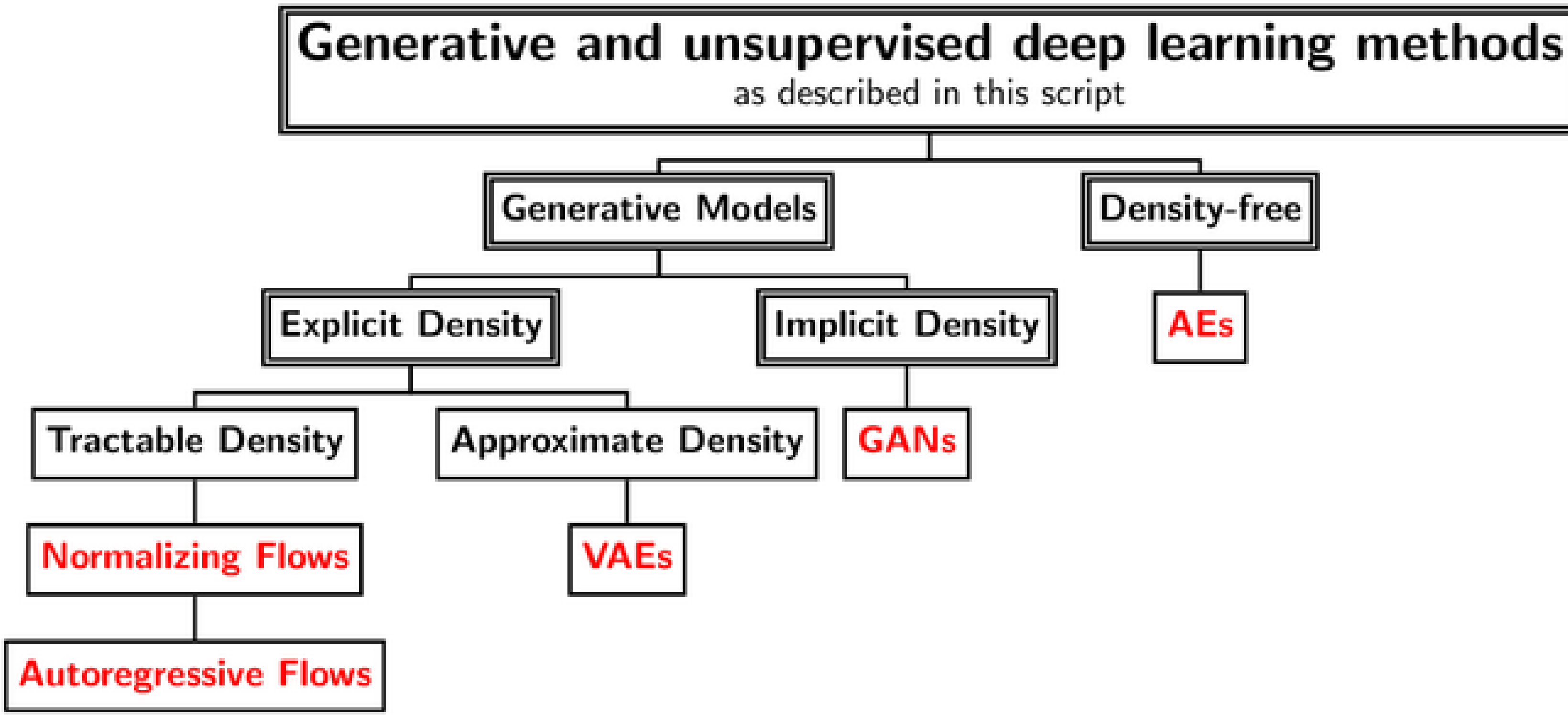
DCGAN

VAE vs GAN images

VAE

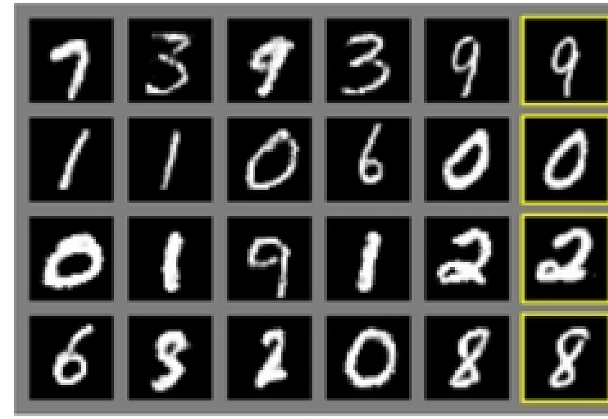
DCGAN

Overview of generative, deep models



12.4. Generative adversarial nets (GANs)

- Suggested in 2014 by Ian Goodfellow
- New type of network and loss, in which two networks play against each other
  - Discriminator network: tries to discriminate between real and generated images
  - Generator network: tries to generate realistic images, to trick the discriminator

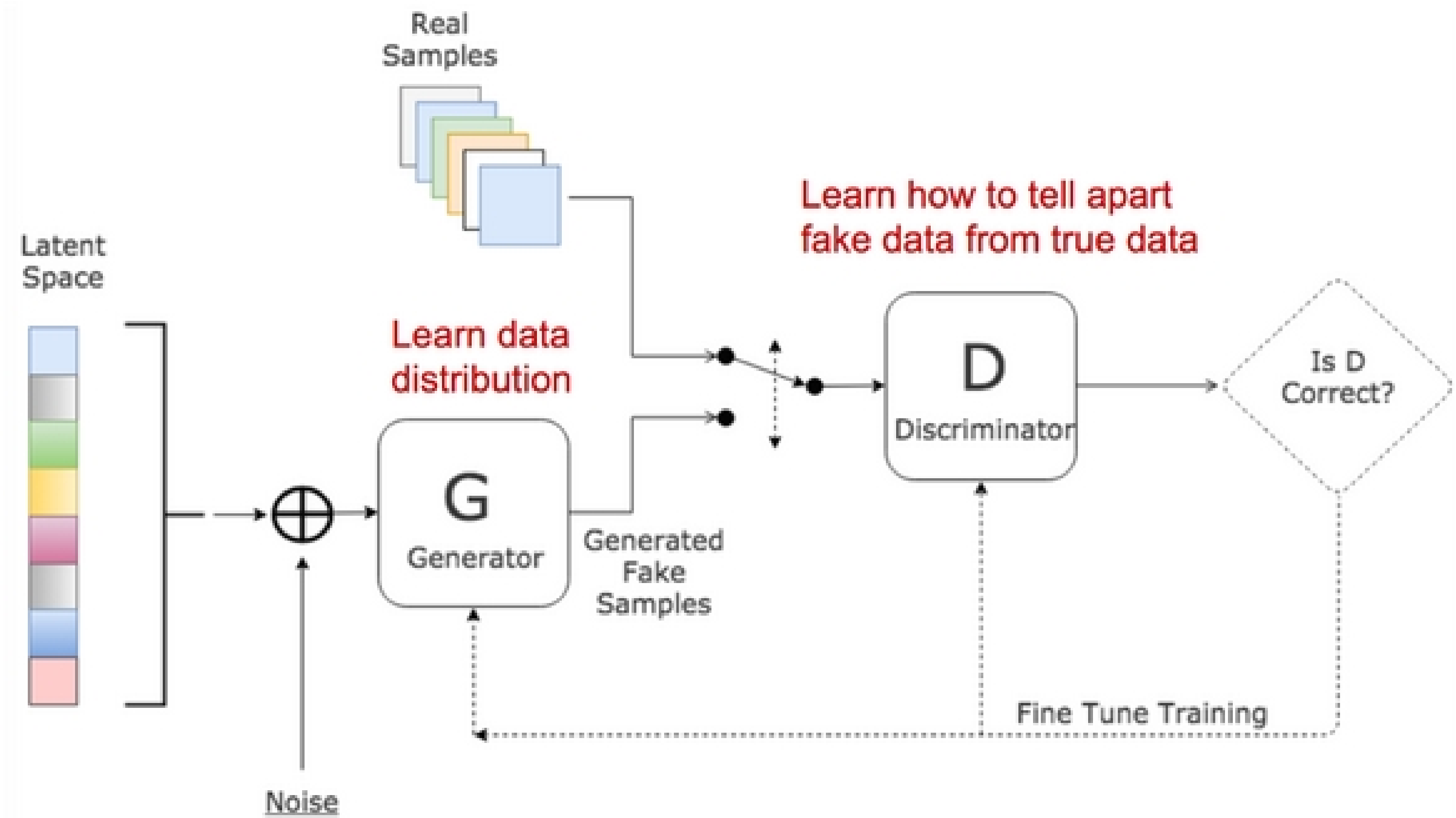


a)

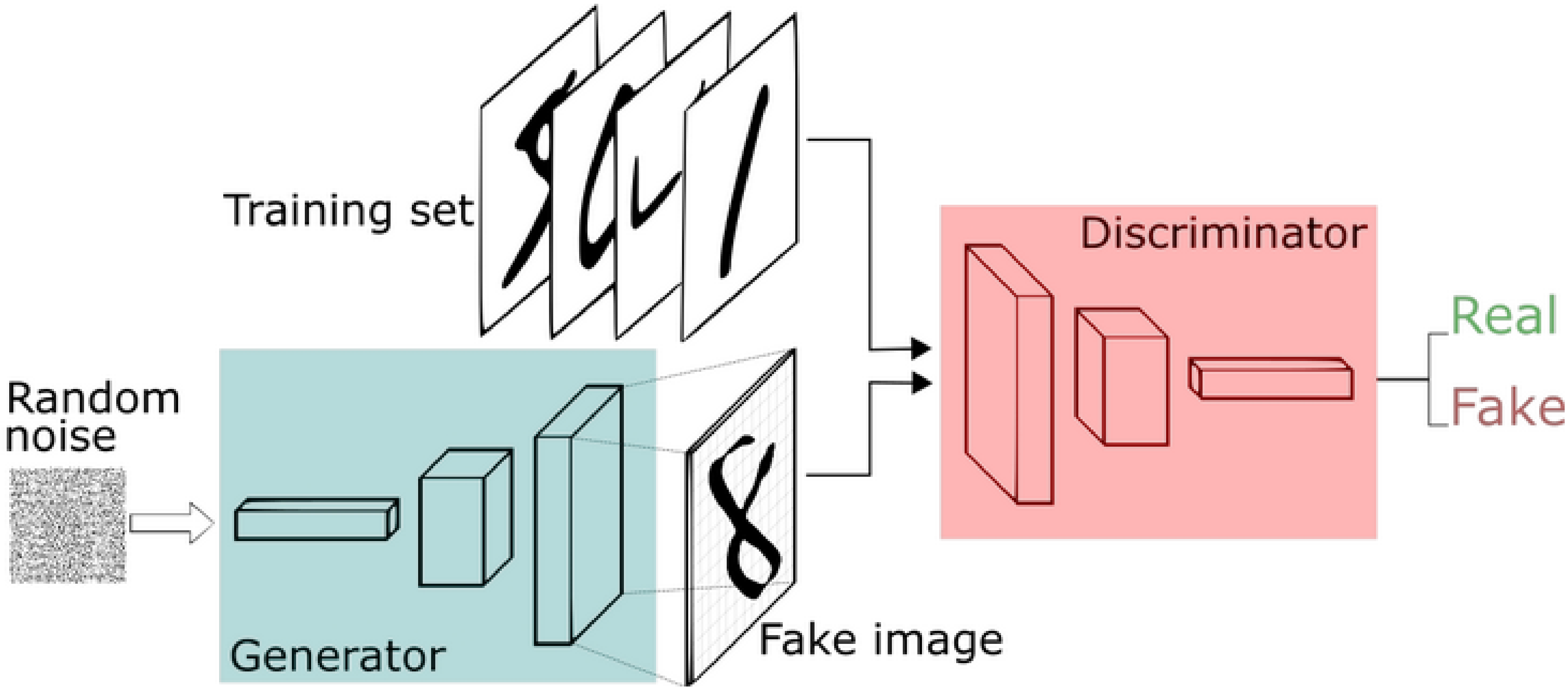
b)

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2014). Generative adversarial nets. In Advances in neural information processing systems (pp. 2672-2680).

The GAN framework



The GAN framework (alternative view)



Objective (loss) function for GANs

$$\begin{aligned}\min_G \max_D \mathcal{L}(D,G) &= \mathbb{E}_{\boldsymbol{x}\sim p_r}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z}\sim p_z}[\log(1 - D(G(\boldsymbol{z})))] \\ &= \mathbb{E}_{\boldsymbol{x}\sim p_r}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{x}\sim p_g}[\log(1 - D(\boldsymbol{x})]\end{aligned}$$

- Two functions have to be optimized
  - Discriminator D
  - Generator G
- Distributions:
  - Random noise $\boldsymbol{z} \sim p_{\boldsymbol{z}}$
  - Real world samples / data distribution $\boldsymbol{x} \sim p_r$
  - Samples from the generator $\boldsymbol{x} \sim p_g$

Progress of images produced by GANs

Further details on GANs

- Pen & paper lecture

The GAN training algorithm

Algorithm 12.1 GAN training

BEGIN GAN training
while Number of epochs $\leq$ N **do**
Sample minibatch of size M from data: $\boldsymbol{x}^1, \ldots \boldsymbol{x}^M \sim p_r$
Sample minibatch of size M of noise: $\boldsymbol{z}^1, \ldots, \boldsymbol{z}^M \sim p_z$
Take a gradient descent step on the generator parameters $\boldsymbol{\theta}$:

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}(D_{\boldsymbol{w}}, G_{\boldsymbol{\theta}}) = \frac{1}{M} \nabla_{\boldsymbol{\theta}} \sum_{i=1}^{M} \log\left(1 - G_{\boldsymbol{\theta}}(D_{\boldsymbol{w}}(\boldsymbol{z}^i))\right)$$

Take a gradient ascent step on the discriminator parameters $\boldsymbol{w}$:

$$\nabla_{\boldsymbol{w}} \mathcal{L}(D_{\boldsymbol{w}}, G_{\boldsymbol{\theta}}) = \frac{1}{M} \nabla_{\boldsymbol{w}} \sum_{i=1}^{M} \log(D_{\boldsymbol{w}}(\boldsymbol{x}^i)) + \log\left(1 - G_{\boldsymbol{\theta}}(D_{\boldsymbol{w}}(\boldsymbol{z}^i))\right)$$

end while
END GAN training